Finite Difference Solution of the One-Group Neutron Diffusion Equation in a Spherical Reactor

Arjun Pawar
25B2286
EN613- Nuclear Reactor Theory

October 14, 2026

Contents

1 Problem Data and Common Formulation **3**

1.1 Material data . . . 3
1.2 Diffusion equation in spherical geometry . . . 3
1.3 Boundary conditions . . . 3
1.4 Uniform grid . . . 3
1.5 Central-difference approximations . . . 3

2 Problem 1: Flux for a Uniform Unit Source (Bare Sphere) **3**

2.1 Problem statement . . . 3
2.2 Analytical solution . . . 4
2.3 (a) Finite difference scheme . . . 4
2.4 (b) Results . . . 5

3 Problem 2: Multiplication Factor k (Bare Sphere) **5**

3.1 Problem statement . . . 5
3.2 Mathematical formulation . . . 6
3.3 (a) Finite difference scheme . . . 6
3.4 (a) Results . . . 6
3.5 (b) Critical value of $\nu\Sigma_f$ . . . 7

4 Problem 3: Sphere with a 5 cm Water Reflector **7**

4.1 Problem statement . . . 7
4.2 Interface conditions . . . 8
4.3 Problem 3(a): Flux for a uniform unit source in the core . . . 8
4.3.1 Analytical solution . . . 8

4.3.2 Finite difference scheme 8
4.3.3 Results 9
4.4 Problem 3(b): k and critical $\nu\Sigma_f$ with the reflector 9
4.4.1 Analytical solution 9
4.4.2 Finite difference scheme (generalized eigenvalue problem) 10
4.4.3 Results 10

5 Conclusions **11**

1 Problem Data and Common Formulation

1.1 Material data

Region	D (cm)	Σ_a (cm^{-1})	$\nu\Sigma_f$ (cm^{-1})
Core ($0 \le r \le 10$ cm)	0.3543	0.121	0.18514
Reflector ($10 \le r \le 15$ cm)	0.16	0.0197	0

Table 1: Material properties. $\nu\Sigma_f$ is used only in Problems 2 and 3(b).

1.2 Diffusion equation in spherical geometry

For a one-group, steady-state problem with spherical symmetry the diffusion equation is

$$-D\,\nabla^2\phi(r) + \Sigma_a\,\phi(r) = S(r), \qquad \nabla^2\phi = \frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{d\phi}{dr}\right) = \frac{d^2\phi}{dr^2} + \frac{2}{r}\frac{d\phi}{dr}. \tag{1}$$

Dividing by D and defining the diffusion length $L^2 = D/\Sigma_a$,

$$\frac{d^2\phi}{dr^2} + \frac{2}{r}\frac{d\phi}{dr} - \frac{1}{L^2}\phi = -\frac{S}{D}. \tag{2}$$

1.3 Boundary conditions

- **Centre:** by symmetry the flux must be finite and smooth, $\left.\dfrac{d\phi}{dr}\right|_{r=0} = 0$.
- **Outer surface:** a vacuum boundary, $\phi(R_{\text{outer}}) = 0$. The outer radius is $R = 10$ cm for the bare core and $R_t = 15$ cm with the reflector.

1.4 Uniform grid

The domain $[0, R_{\text{outer}}]$ is divided into N equal intervals of width $\Delta r = R_{\text{outer}}/N$, with nodes $r_i = i\,\Delta r$, $i = 0, 1, \ldots, N$, and $\phi_i \approx \phi(r_i)$. The boundary condition gives $\phi_N = 0$, so the unknowns are $\phi_0, \ldots, \phi_{N-1}$.

1.5 Central-difference approximations

At an interior node i the derivatives are replaced by second-order central differences,

$$\left.\frac{d^2\phi}{dr^2}\right|_i \approx \frac{\phi_{i+1} - 2\phi_i + \phi_{i-1}}{\Delta r^2}, \qquad \left.\frac{d\phi}{dr}\right|_i \approx \frac{\phi_{i+1} - \phi_{i-1}}{2\Delta r}. \tag{3}$$

2 Problem 1: Flux for a Uniform Unit Source (Bare Sphere)

2.1 Problem statement

Find the flux distribution in a bare sphere of radius $R = 10$ cm with a uniform unit source $S = 1$ neutron cm^{-3}s^{-1}, $\Sigma_a = 0.121$ cm^{-1} and $D = 0.3543$ cm.

2.2 Analytical solution

Equation (2) with constant S has the particular solution $\phi_p = S/\Sigma_a$. For the homogeneous part substitute $u(r) = r\phi(r)$: since $\nabla^2\phi = \frac{1}{r}\frac{d^2u}{dr^2}$, the homogeneous equation becomes $u'' - u/L^2 = 0$, so $u = A\sinh(r/L) + B\cosh(r/L)$ and

$$\phi(r) = \frac{S}{\Sigma_a} + A\,\frac{\sinh(r/L)}{r} + B\,\frac{\cosh(r/L)}{r}. \tag{4}$$

Finiteness of the flux at $r = 0$ forces $B = 0$. The vacuum condition $\phi(R) = 0$ gives $A = -\dfrac{S}{\Sigma_a}\dfrac{R}{\sinh(R/L)}$, hence

$$\boxed{\phi(r) = \frac{S}{\Sigma_a}\left[1 - \frac{R}{r}\,\frac{\sinh(r/L)}{\sinh(R/L)}\right]} \tag{5}$$

2.3 (a) Finite difference scheme

Interior nodes ($1 \le i \le N-1$). Inserting (3) into (2) gives

$$\frac{\phi_{i+1} - 2\phi_i + \phi_{i-1}}{\Delta r^2} + \frac{1}{r_i}\,\frac{\phi_{i+1} - \phi_{i-1}}{\Delta r} - \frac{\Sigma_a}{D}\phi_i = -\frac{S}{D}, \tag{6}$$

and collecting terms,

$$\underbrace{\left(\frac{1}{\Delta r^2} - \frac{1}{r_i \Delta r}\right)}_{a_i}\phi_{i-1} \;-\; \underbrace{\left(\frac{2}{\Delta r^2} + \frac{\Sigma_a}{D}\right)}_{b_i}\phi_i \;+\; \underbrace{\left(\frac{1}{\Delta r^2} + \frac{1}{r_i \Delta r}\right)}_{c_i}\phi_{i+1} = -\frac{S}{D}. \tag{7}$$

Centre node ($i = 0$). The term $\frac{2}{r}\phi'$ is singular at $r = 0$, so the limit is taken using L'Hôpital's rule: since $\phi'(0) = 0$, $\lim_{r\to 0}\frac{\phi'}{r} = \phi''(0)$ and therefore

$$\nabla^2\phi\big|_{r=0} = \phi''(0) + 2\phi''(0) = 3\,\phi''(0).$$

Using the symmetry condition, a fictitious node $\phi_{-1} = \phi_1$ gives $\phi''(0) \approx 2(\phi_1 - \phi_0)/\Delta r^2$, so

$$\left(-\frac{6}{\Delta r^2} - \frac{\Sigma_a}{D}\right)\phi_0 + \frac{6}{\Delta r^2}\phi_1 = -\frac{S}{D}. \tag{8}$$

Outer boundary ($i = N$). $\phi_N = 0$.

Matrix form. Equations (7), (8) and the boundary condition form a tridiagonal linear system (which luckily i have been taught recently in one of my core courses ME225- Numerical Analysis to solve)$\mathbf{A}\boldsymbol{\phi} = \mathbf{c}$ of size $(N+1)\times(N+1)$:

$$\begin{bmatrix} -\frac{6}{\Delta r^2} - \frac{\Sigma_a}{D} & \frac{6}{\Delta r^2} & & & \\ a_1 & -b & c_1 & & \\ & \ddots & \ddots & \ddots & \\ & & a_{N-1} & -b & c_{N-1} \\ & & & 0 & 1 \end{bmatrix} \begin{bmatrix} \phi_0 \\ \phi_1 \\ \vdots \\ \phi_{N-1} \\ \phi_N \end{bmatrix} = \begin{bmatrix} -S/D \\ -S/D \\ \vdots \\ -S/D \\ 0 \end{bmatrix}, \tag{9}$$

which is solved directly (`numpy.linalg.solve`).

2.4 (b) Results

The problem was solved for $N = 5, 10, 15, 20$ ($\Delta r = 2, 1, 0.667, 0.5$ cm) and compared with Eq. (5).

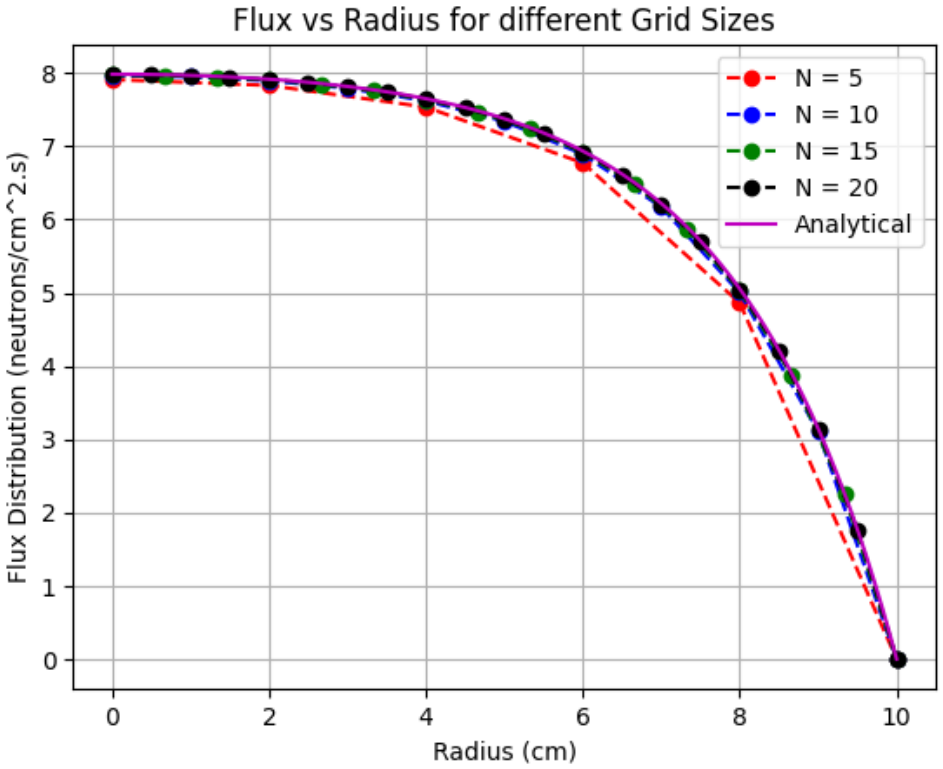


Figure 1: Problem 1: flux versus radius for $N = 5, 10, 15, 20$ compared with the analytical solution.

N	Δr (cm)	Max. absolute error
5	2.000	0.1915
10	1.000	0.0518
15	0.667	0.0234
20	0.500	0.0132

Table 2: Maximum absolute error with respect to the analytical flux (nodes $i \geq 1$).

Discussion. The error falls by roughly a factor of 4 when the grid spacing is halved (0.0518 $\rightarrow$ 0.0132 from $N = 10$ to $N = 20$), confirming the expected second-order accuracy $O(\Delta r^2)$ of the central-difference scheme.

3 Problem 2: Multiplication Factor k (Bare Sphere)

3.1 Problem statement

With no external source and $\nu\Sigma_f = 0.18514$ cm^{-1}, find k for the same sphere.

3.2 Mathematical formulation

Without an external source the fission neutrons, scaled by $1/k$, form the source:

$$-D\nabla^2\phi + \Sigma_a\phi = \frac{1}{k}\,\nu\Sigma_f\,\phi. \tag{10}$$

This is an eigenvalue problem. The fundamental (positive) flux and the largest k are of physical interest.

Analytical solution. Writing (10) as $\nabla^2\phi + B^2\phi = 0$ with the material buckling $B^2 = (\nu\Sigma_f/k - \Sigma_a)/D$, the finite solution at the centre is $\phi = \sin(Br)/r$. The condition $\phi(R) = 0$ requires $BR = n\pi$; the fundamental mode is $B = \pi/R$:

$$\phi(r) = \frac{\sin(\pi r/R)}{r}, \qquad \boxed{k = \frac{\nu\Sigma_f}{\Sigma_a + D\,(\pi/R)^2}} \tag{11}$$

which for the given data gives $k_{\text{analytical}} = 1.18704$.

3.3 (a) Finite difference scheme

The spatial discretization is identical to Problem 1 (Eqs. (7) and (8)), but without the source term. Let $\mathbf{M}$ be the $N \times N$ matrix obtained from the rows $i = 0, \ldots, N-1$ (the column for the known value $\phi_N = 0$ is dropped). The discrete form of (10), divided by D, is

$$\mathbf{M}\boldsymbol{\phi} = -\frac{\nu\Sigma_f}{k\,D}\,\boldsymbol{\phi} \quad\Longleftrightarrow\quad (-\mathbf{M})\boldsymbol{\phi} = \lambda\boldsymbol{\phi}, \qquad \lambda = \frac{\nu\Sigma_f}{k\,D}. \tag{12}$$

The fundamental mode corresponds to the *smallest* eigenvalue $\lambda_{\min}$ of $-\mathbf{M}$ (the one with a non-negative flux of one sign), so

$$k = \frac{\nu\Sigma_f}{D\,\lambda_{\min}}. \tag{13}$$

The eigenvector is normalized to its maximum value, $\phi_i \leftarrow \phi_i / \max_j \phi_j$, because the flux level in an eigenvalue problem is arbitrary.

3.4 (a) Results

The problem was solved for $N = 5, 8, 12, 16$.

N	k (numerical)	k (analytical)	$\lvert k_{\text{num}} - k_{\text{exact}}\rvert$
5	1.19574	1.18704	0.00870
8	1.19045	1.18704	0.00341
12	1.18856	1.18704	0.00152
16	1.18789	1.18704	0.00085

Table 3: Multiplication factor for different grid sizes (bare sphere).

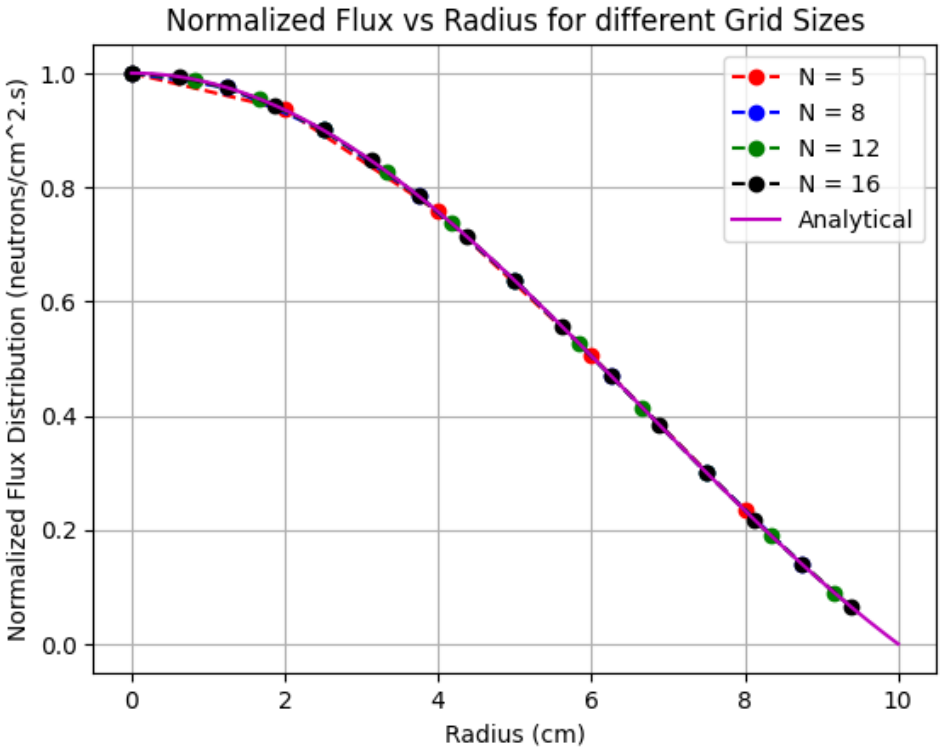


Figure 2: Problem 2: normalized flux for $N = 5, 8, 12, 16$ compared with the analytical normalized flux $\sin(\pi r/R)/r$.

3.5 (b) Critical value of $\nu\Sigma_f$

In Eq. (10), k is directly proportional to $\nu\Sigma_f$. The reactor is critical when $k = 1$. The value of $\nu\Sigma_f$ was found by bisection on the finest grid ($N = 16$): start from the bracket [0.10, 0.25], evaluate k at the midpoint, and move the lower end up if $k < 1$ or the upper end down if $k > 1$. Because $k = \nu\Sigma_f/(D\lambda_{\min})$, the result is also $\nu\Sigma_f^{\text{crit}} = D\lambda_{\min}$, which provides a check. The analytical value is $\nu\Sigma_f^{\text{crit}} = \Sigma_a + D(\pi/R)^2 = 0.15597$ cm^{-1}.

	Numerical ($N = 16$)	Analytical
Critical $\nu\Sigma_f$ (cm^{-1})	0.15586	0.15597

Table 4: Critical value of $\nu\Sigma_f$ for the bare sphere.

4 Problem 3: Sphere with a 5 cm Water Reflector

4.1 Problem statement

The core ($0 \le r \le R_c = 10$ cm) is surrounded by a reflector of thickness $T = 5$ cm ($R_c \le r \le R_t = 15$ cm), with $D_r = 0.16$ cm, $\Sigma_{a,r} = 0.0197$ cm^{-1}, and no source (or fission) in the reflector. Problems 1 and 2 are repeated.

4.2 Interface conditions

The diffusion equation is now solved with piecewise constant coefficients. At the core–reflector interface $r = R_c$ two physical conditions hold:

$$\phi\big|_{R_c^-} = \phi\big|_{R_c^+} \quad \text{(continuity of flux)}, \tag{14}$$

$$D_c\,\phi'\big|_{R_c^-} = D_r\,\phi'\big|_{R_c^+} \quad \text{(continuity of current)}. \tag{15}$$

The outer boundary is $\phi(R_t) = 0$ and at the centre $\phi'(0) = 0$.

4.3 Problem 3(a): Flux for a uniform unit source in the core

4.3.1 Analytical solution

Let $L_c^2 = D_c/\Sigma_{a,c}$ and $L_r^2 = D_r/\Sigma_{a,r}$. The solution regular at the centre, and vanishing at R_t, is

$$\phi(r) = \begin{cases} \dfrac{S}{\Sigma_{a,c}} + A\,\dfrac{\sinh(r/L_c)}{r}, & 0 \le r \le R_c, \\[2ex] C\,\dfrac{\sinh\big((R_t - r)/L_r\big)}{r}, & R_c \le r \le R_t. \end{cases} \tag{16}$$

Using $\frac{d}{dr}\left[\frac{\sinh(r/L)}{r}\right] = \frac{\cosh(r/L)}{L\,r} - \frac{\sinh(r/L)}{r^2}$, the two interface conditions give a 2×2 linear system for A and C:

$$\begin{bmatrix} \dfrac{\sinh(R_c/L_c)}{R_c} & -\dfrac{\sinh(T/L_r)}{R_c} \\[2ex] D_c\left[\dfrac{\cosh(R_c/L_c)}{L_c R_c} - \dfrac{\sinh(R_c/L_c)}{R_c^2}\right] & D_r\left[\dfrac{\cosh(T/L_r)}{L_r R_c} + \dfrac{\sinh(T/L_r)}{R_c^2}\right] \end{bmatrix} \begin{bmatrix} A \\ C \end{bmatrix} = \begin{bmatrix} -S/\Sigma_{a,c} \\ 0 \end{bmatrix}. \tag{17}$$

4.3.2 Finite difference scheme

The grid covers $[0, R_t]$ with $\Delta r = R_t/N$. The interface R_c is placed *exactly on a node*, $i_c = 2N/3$, so N **must be a multiple of 3**.

Nodes inside the core ($0 \le i < i_c$) and inside the reflector ($i_c < i < N$). The same equations as in Problem 1 are used, with $(D_c, \Sigma_{a,c}, S)$ in the core and $(D_r, \Sigma_{a,r}, 0)$ in the reflector, including the centre equation (8).

Interface node ($i = i_c$). A direct central difference is not valid because D is discontinuous. Instead, the diffusion equation multiplied by r^2 is integrated over the control volume $[R_c - \Delta r/2,\ R_c + \Delta r/2]$:

$$-\Big[D\,r^2\phi'\Big]_{R_c-\Delta r/2}^{R_c+\Delta r/2} + \int \Sigma_a\,\phi\,r^2\,dr = \int S\,r^2\,dr.$$

Approximating the face derivatives by $(\phi_{i_c} - \phi_{i_c-1})/\Delta r$ and $(\phi_{i_c+1} - \phi_{i_c})/\Delta r$ gives, with $r_\mp = R_c \mp \Delta r/2$,

$$\frac{D_c\,r_-^2}{\Delta r}\,\phi_{i_c-1} \;-\; \left[\frac{D_c\,r_-^2}{\Delta r} + \frac{D_r\,r_+^2}{\Delta r} + \Sigma_{a,c}V_c + \Sigma_{a,r}V_r\right]\phi_{i_c} \;+\; \frac{D_r\,r_+^2}{\Delta r}\,\phi_{i_c+1} \;=\; -S\,V_c, \tag{18}$$

where the core and reflector volume factors are

$$V_c = \frac{R_c^3 - r_-^3}{3}, \qquad V_r = \frac{r_+^3 - R_c^3}{3}. \tag{19}$$

This form automatically conserves neutrons across the interface and enforces continuity of $D\phi'$.

Outer boundary. $\phi_N = 0$. The resulting system $\mathbf{A}\boldsymbol{\phi} = \mathbf{c}$ is again solved directly.

4.3.3 Results

The problem was solved for $N = 6, 12, 18, 21$.

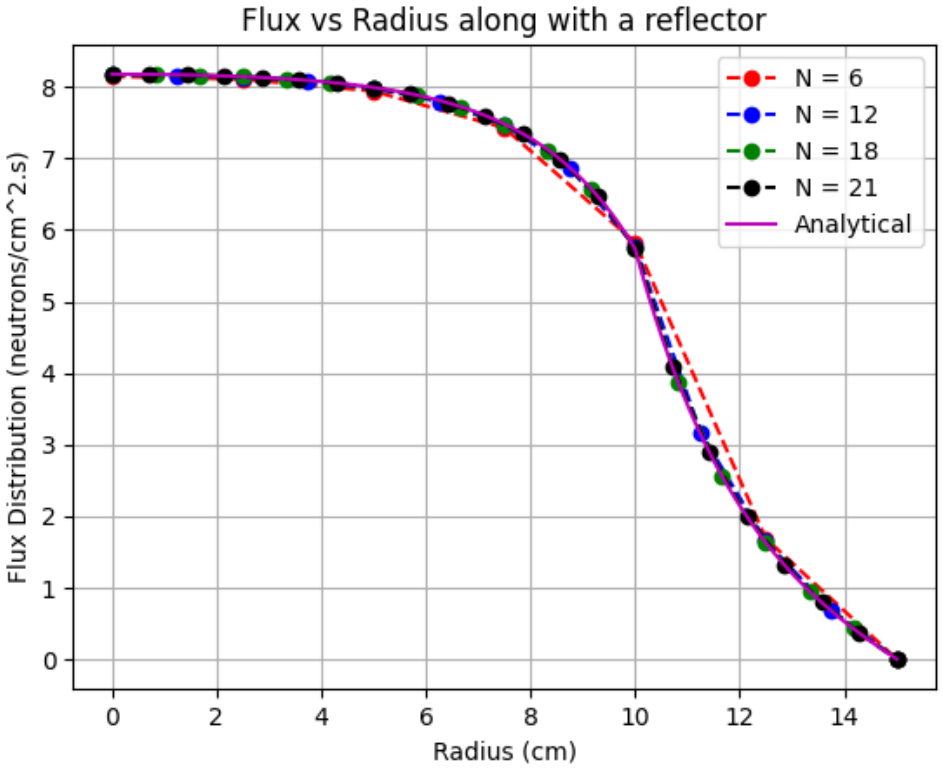


Figure 3: Problem 3(a): flux versus radius with a 5 cm reflector for $N = 6, 12, 18, 21$ compared with the analytical solution. The change of slope at $r = 10$ cm is the core–reflector interface.

4.4 Problem 3(b): k and critical $\nu\Sigma_f$ with the reflector

4.4.1 Analytical solution

Fission occurs only in the core, so

$$-D_c\nabla^2\phi + \Sigma_{a,c}\phi = \frac{\nu\Sigma_f}{k}\phi \qquad (r < R_c), \tag{20}$$

$$-D_r\nabla^2\phi + \Sigma_{a,r}\phi = 0 \qquad (R_c < r < R_t). \tag{21}$$

With $B^2 = (\nu\Sigma_f/k - \Sigma_{a,c})/D_c$ and $\kappa^2 = \Sigma_{a,r}/D_r = 1/L_r^2$, the solution is

$$\phi(r) = \begin{cases} \dfrac{\sin(Br)}{r}, & r \le R_c, \\[2ex] \dfrac{\sin(BR_c)}{\sinh(\kappa T)}\,\dfrac{\sinh\big(\kappa(R_t - r)\big)}{r}, & r \ge R_c, \end{cases} \tag{22}$$

where the reflector constant has already been fixed by flux continuity at R_c. Continuity of $D\phi'$ at R_c, using $\phi'/\phi = B\cot(BR_c) - 1/R_c$ in the core and $\phi'/\phi = -\kappa\coth(\kappa T) - 1/R_c$ in the reflector, gives the transcendental equation for B:

$$D_c\left[B\cot(BR_c) - \frac{1}{R_c}\right] + D_r\left[\kappa\coth(\kappa T) + \frac{1}{R_c}\right] = 0, \qquad \frac{\pi}{2R_c} < B < \frac{\pi}{R_c}. \tag{23}$$

This is solved by bisection, and then

$$\boxed{k = \frac{\nu\Sigma_f}{\Sigma_{a,c} + D_c B^2}} \tag{24}$$

The root lies below π/R_c: the reflector lowers the effective buckling of the system compared with the bare core, and the equation gives $B = 0.20716$ cm^{-1} and $k_{\text{analytical}} = 1.35927$.

4.4.2 Finite difference scheme (generalized eigenvalue problem)

The matrix $\mathbf{M}$ is the one from Problem 3(a) (rows $0, \ldots, N-1$) without the source. Because the fission source exists only in the core and each matrix row was divided by a different factor, the problem is a *generalized* eigenvalue problem and the formula $k = \nu\Sigma_f/(D\lambda)$ of Problem 2 can no longer be used.

Let s_i be the factor by which the equation of node i was scaled:

$$s_i = \begin{cases} 1/D_c & \text{core nodes } (i < i_c), \\ V_c & \text{interface node } (i = i_c), \\ 0 & \text{reflector nodes } (i > i_c). \end{cases}$$

(These are the same factors that multiply the source S in Problem 3(a).) The source $\nu\Sigma_f\phi_i/k$ must be scaled in the same way, so that

$$(-\mathbf{M})\,\boldsymbol{\phi} = \frac{1}{k}\,\mathbf{F}\,\boldsymbol{\phi}, \qquad \mathbf{F} = \operatorname{diag}(\nu\Sigma_f\, s_0,\ \nu\Sigma_f\, s_1,\ \ldots,\ \nu\Sigma_f\, s_{N-1})\,. \tag{25}$$

Multiplying by $(-\mathbf{M})^{-1}$,

$$\big[(-\mathbf{M})^{-1}\mathbf{F}\big]\boldsymbol{\phi} = k\,\boldsymbol{\phi}, \tag{26}$$

so k is the *largest* eigenvalue of $(-\mathbf{M})^{-1}\mathbf{F}$ and its eigenvector is the fundamental flux, normalized to a maximum of 1. (Using the smallest eigenvalue, as in Problem 2, would select a higher harmonic.) This requires N to be a multiple of 3.

4.4.3 Results

N	k (numerical)	k (analytical)	$\lvert k_{\text{num}} - k_{\text{exact}}\rvert$
6	1.34740	1.35927	0.01187
9	1.35363	1.35927	0.00564
12	1.35602	1.35927	0.00325
18	1.35779	1.35927	0.00148

Table 5: Multiplication factor with reflector for different grid sizes.

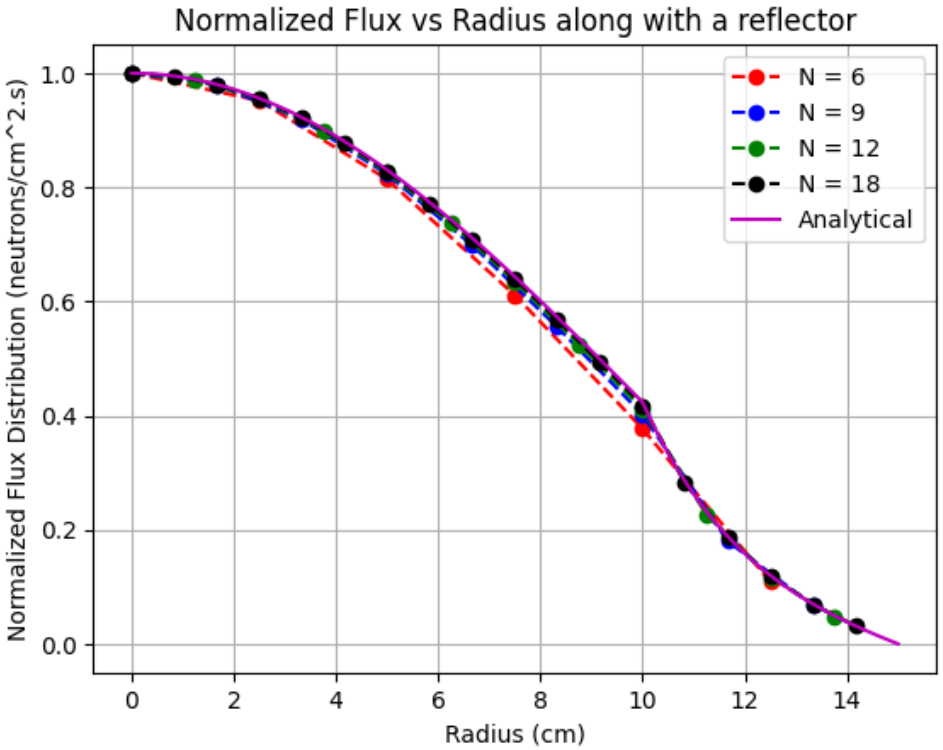


Figure 4: Problem 3(b): normalized flux with a 5 cm reflector for $N = 6, 9, 12, 18$ compared with the analytical normalized flux.

Critical $\nu\Sigma_f$. Using the finest grid ($N = 18$) and the same bisection on $\nu\Sigma_f$ as before:

	Numerical ($N = 18$)	Analytical
Critical $\nu\Sigma_f$ (cm^{-1})	0.13635	0.13621

Table 6: Critical $\nu\Sigma_f$ with the reflector. The analytical value is $\Sigma_{a,c} + D_c B^2$.

5 Conclusions

- The spherical diffusion equation was discretized with second-order central differences; the singularity at $r = 0$ was handled through the symmetry condition giving $\nabla^2\phi \approx 6(\phi_1 - \phi_0)/\Delta r^2$.
- In all three problems the numerical solution converges to the analytical solution as the grid is refined, with an error that decreases roughly as Δr^2.
- The reflector lowers the critical $\nu\Sigma_f$ from 0.15597 to 0.13621 cm^{-1} (about 13%) and increases k from 1.187 to 1.359 for the same fuel, by returning leaked neutrons to the core.
- With a reflector the interface must be treated with a flux-conservative (finite volume) equation, and k must be found from a generalized eigenvalue problem because fission is present in the core only.